



The role of curcumin in modulating circular RNAs and long non-coding RNAs in cancer

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Abstract

Cancer is one of the primary causes of human disease and death, with high morbidity and mortality rates. Chemotherapy, one of the most common therapeutic techniques, functions through a variety of mechanisms, including the production of apoptosis and the prevention of tumor development. Herbal medicine has been the subject of numerous investigations due to its potential as a valuable source of innovative anti-cancer products that target multiple protein targets and cancer cell genomes. Curcumin, a polyphenol that is the major bioactive ingredient of turmeric, exhibits pharmacological and biological efficacy with antioxidant, anti-inflammatory, anticancer, cardioprotective, neuroprotective, and hypoglycemic activity in humans and animals. Recent research suggests that curcumin changes noncoding RNA (ncRNA), such as long noncoding RNAs (lncRNAs) and circular RNAs (circRNAs), in various types of cancers. Both circRNAs and lncRNAs are ncRNAs that can epigenetically modulate the expression of multiple genes via post-transcriptional regulation. In this study, we outline curcumin's activities in modulating signaling pathways and ncRNAs in various malignancies. We also described curcumin's regulatory function, which involves blocking carcinogenic lncRNAs and circRNAs while increasing tumor-suppressive ones. Furthermore, we intend to demonstrate how ncRNAs and signaling pathways interact with each other across regulatory boundaries to gain a better understanding of how curcumin fights cancer and create a framework for its potential future therapeutic uses.

Keywords Cancer · Curcumin · lncRNAs · circRNAs · Signaling pathways

Introduction

Due to its medicinal properties, people have utilized curcumin's unique combination of features for thousands of years. Numerous articles today explore the diverse activities of curcumin, including its antioxidant, anti-inflammatory, antibacterial, anti-apoptotic, anticancer, neuroprotective, and anti-diabetic properties [1, 2]. Therefore, curcumin may serve as a potential therapeutic agent for various chronic and inflammatory diseases, including diabetes, neurological disorders, cardiovascular diseases, and cancers [1, 3–5]. Researchers have discovered that curcumin possesses both preventative and therapeutic properties against certain malignancies [6].

Curcumin has been shown to suppress tumor formation and progression by blocking the activity of angiogenesis, lowering the changes in matrix metalloproteases (MMPs) and adhesive molecules on the cell surface, stopping tumor cell growth, upregulating tumor suppressor genes like P53, and inducing apoptosis [7, 8].

Recent research indicates that curcumin is an epigenetic regulator of noncoding RNAs in several malignancies [9]. Noncoding RNAs are generated from the genome but cannot encode proteins. They are categorized into two types based on their length threshold of 200 nucleotides [10]. Long noncoding RNAs contribute to tumor growth through processes such as mRNA regulation. Research has demonstrated that long noncoding RNAs influence tumor growth through epigenetic regulation, DNA destruction, cell cycle control, and chromosomal instability, highlighting their therapeutic value for malignancies [11, 12].

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Recent research has shown that some long noncoding RNAs can form circular RNAs and act as sponges, attracting microRNAs or transcription factors to influence the expression of target genes [13]. Circular RNAs' circular structure makes them more stable than linear noncoding RNAs, which are resistant to most RNA decay mechanisms [14]. They exhibit tissue- and cell-specific expression patterns, and their synthesis is regulated by a type of alternative splicing called backsplicing [15]. Although long noncoding RNAs and circular RNAs are relatively rare, their expression varies significantly among different cell types, tissues, developmental stages, and disease states. Recent independent investigations have demonstrated that the deregulation of long noncoding RNAs and circular RNAs is important in cancer development, progression, and angiogenesis [16, 17]. Curcumin can inhibit tumor proliferation, angiogenesis, epithelial-mesenchymal transition (EMT), invasion and metastasis by regulating tumor-related noncoding RNA expression [18–20]. The regulation of these noncoding RNAs targeted the modulation of mRNAs and modified various cancer signaling pathways [21]. However, the interaction between their activities and curcumin's components in tumor progression remains largely unknown. Curcumin's ability to suppress tumors in a variety of tumor types by influencing long noncoding RNAs is gaining more and more evidence.

The aim of this study is to investigate the effects of curcumin on the regulation and expression of long noncoding RNAs and circular RNAs. We intend to learn more about how curcumin affects these noncoding RNA structures at the molecular level so that we can better understand how it might be used as a medicine to change gene expression and cellular functions. This research will contribute to the growing body of knowledge regarding the role of curcumin in RNA-mediated regulatory pathways and its implications for various biological processes in cancer.

Curcumin: molecular information

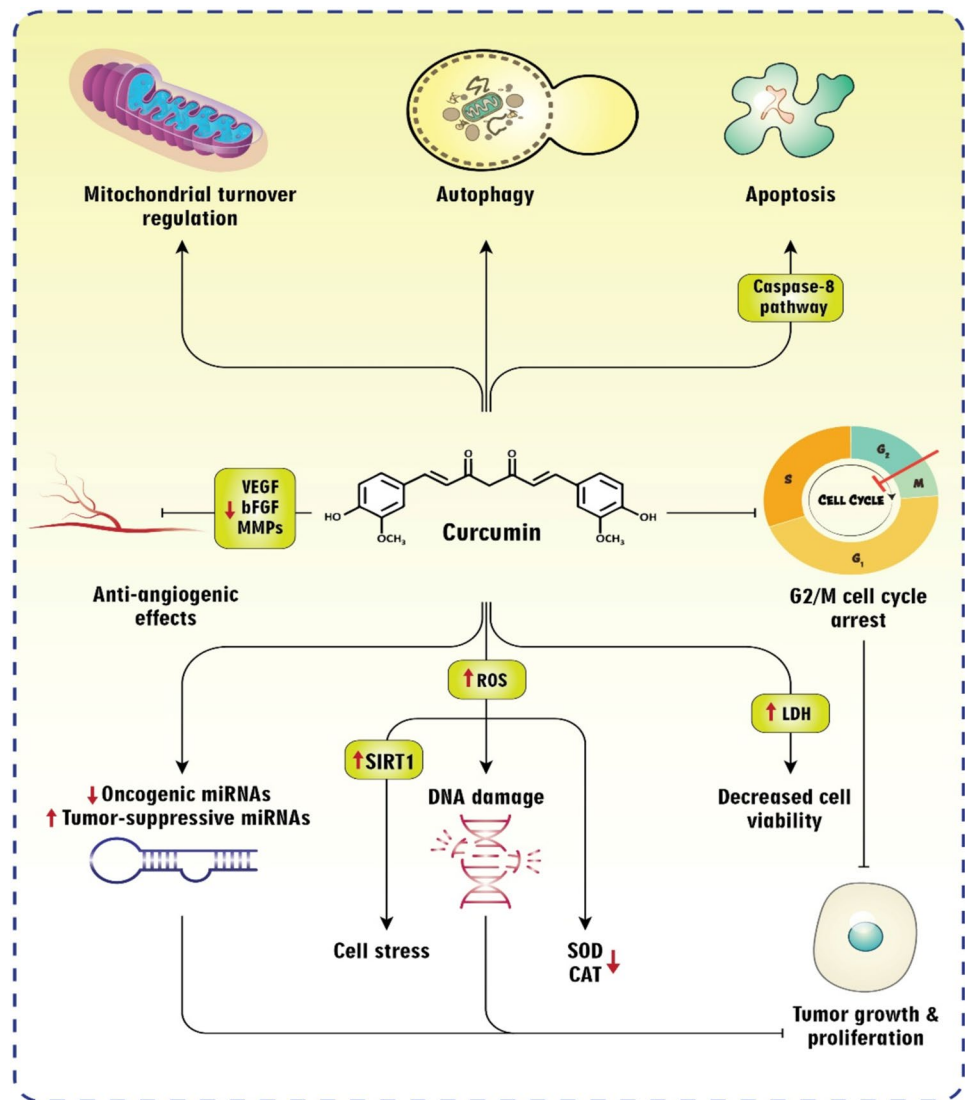
In 1815, the efforts of Vogel and Pelletier led to the discovery of a yellowish substance, curcumin, extracted from a tropical plant in the ginger family known as *Curcuma longa* or turmeric. This plant is primarily found in India, Indonesia, and South Asia [22]. The isolation of curcumin, primarily through ethanol extraction from *Curcuma longa* L., yields the yellow, bioactive, polyphenolic, and hydrophobic pigments that constitute 2–8% of turmeric. Curcumin is used not only as a food ingredient and coloring agent but also as a herbal medication, particularly in Southeast Asia and China, due to its anti-inflammatory, cell growth-regulating, and antioxidant properties [23, 24]. Turmeric has three kinds of curcuminoids: curcumin, bisdemethoxycurcumin, and demethoxycurcumin. It also has volatile oils (turmerone, zingiberone, and atlantone), proteins, sugars, and resins.

Among these curcuminoids, curcumin is notable for its ability to withstand the acidic pH of the stomach, although it is not water-soluble [25].

Milobedska first elucidated the chemical structure and synthesis of curcumin in 1910, followed by Lampe et al. in 1913 [26, 27]. In natural and acidic conditions, as well as in its solid phase, curcumin exists in its ketonic form. Its enolic form generally appears at an elevated pH of 8 (Fig. 1). Although curcumin is not soluble in water with a natural or acidic pH and degrades within 30 min at a basic pH, it readily dissolves in solvents such as acetone, dimethyl sulfoxide, ethanol, and methanol [28]. The enolic form of curcumin allows its presence in solution. Spectrophotometric experiments have shown that the highest absorption of curcumin (λ_{max}) in methanol occurs at 430 nm. One type of pure curcumin is 1,7-bis-[4-hydroxy-3-methoxyphenyl]-1,6-heptadiene-3,5-dione. It has a molecular weight of 368.37 g/mol and is made up of two ferulic acid residues connected by a methylene bridge [29]. Curcumin turns red at pH levels greater than 7 and yellow at pH levels between 2.5 and 7. At a pH of 8, curcumin acts as an electron donor, which contributes to its antioxidant and scavenging properties. Between pH levels of 3 and 7, curcumin has a remarkable potential to donate hydrogen atoms [30].

Turmeric, the primary source of curcumin, is widely used in Asia for its healing properties. Traditionally, Indians have used turmeric to treat various pathological conditions such as rheumatoid arthritis, asthma, sinusitis, anorexia, and diabetic wound healing. The first documented use of curcumin in treating human diseases occurred in 1937 [31]. Thanks to its anti-inflammatory and antioxidant properties, curcumin has become an iconic candidate for the treatment and prevention of cancer, inflammatory diseases, cardiovascular diseases, neurodegenerative disorders, hepatic lesions, AIDS, cognitive impairments like Alzheimer's disease, diabetes, and many other chronic conditions [32]. Researchers have extensively studied curcumin's natural compounds over the past decades, revealing its potential for treating diabetes and obesity. For instance, by activating miR-206, curcumin helps protect against impairments caused by fructose-induced insulin signaling, and it also lowers cholesterol levels [33]. Human studies documented the antibacterial effects of curcumin were documented in 1949, and its ability to lower blood sugar levels was recorded in 1972 [34]. It is worth mentioning that curcumin can reduce insulin resistance, improve β -cell function, prevent β -cell death, and delay the development of type 2 diabetes mellitus (T2DM). However, despite these benefits, curcumin presents a mixed blessing due to its poor bioavailability. Specialists remain concerned about this issue because curcumin undergoes rapid metabolism, poor absorption, and quick elimination, which reduces its concentration in the systemic circulation [35]. To improve

Fig. 1 The mechanisms of action of curcumin in cancer development



its pharmacokinetic properties and increase its bioavailability, various methods have been developed and tested. To do this, different types of pharmaceutical formulations, using curcumin analogues, liposomal formulations, curcumin nanoparticles, and complexes with phospholipids [36].

Curcumin as anti-cancer agent

Curcumin's potential as a therapeutic agent lies in its ability to regulate mitochondrial turnover, autophagy, and apoptosis. This regulation is crucial for maintaining cellular homeostasis and has significant implications for treating various diseases, including cancer. Curcumin can induce G2/M cell cycle arrest, a critical checkpoint where cells repair DNA damage before entering mitosis. It also plays a role in inducing autophagy, a process involving the degradation and recycling of cellular components, particularly damaged

organelles like mitochondria (mitophagy). Furthermore, curcumin impacts apoptosis, the programmed cell death essential for removing unhealthy cells. Additionally, curcumin modulates molecular signaling pathways, inhibiting cancer cell invasion and metastasis and enhancing the efficacy of existing chemotherapeutics [37]. Curcumin inhibits angiogenesis in human intestinal microvascular endothelial cells by suppressing the activation of COX-2 and MAPK induced by vascular endothelial growth factor (VEGF). This suggests that curcumin has antiangiogenic properties, which could be beneficial in treating chronic inflammation and cancer [38]. Curcumin can stimulate angiogenesis in human placental trophoblasts during the first-trimester by enhancing the expression of VEGF. It also inhibits the expression and activity of key angiogenic factors such as VEGF, basic fibroblast growth factor (bFGF), and MMPs. By downregulating these pro-angiogenic factors, curcumin effectively inhibits the proliferation, migration, and tube

formation of endothelial cells, which are crucial steps in the angiogenic process [39]. Curcuminoids can induce DNA damage in cancer cells by generating reactive oxygen species (ROS) through interactions with intracellular copper. The use of a copper chelator and ROS scavengers significantly inhibits this reaction, highlighting the role of copper in the process. Additionally, copper-supplemented medium makes normal breast epithelial cells sensitive to curcumin-mediated cell death. Curcuminoids may help explain why some cells die, like cancer cells, and how this copper-mediated, ROS-induced mechanism works [40].

The study “Curcumin Modulates Oxidative Stress, Fibrosis, and Apoptosis in Drug-Resistant Cancer Cell Lines” investigates the effects of curcumin on cancer cells that have developed resistance to chemotherapy. The findings indicate that curcumin substantially lowered the development of both sensitive and resistant cancer cell lines. This was shown by lower cell proliferation and higher levels of lactate dehydrogenase (LDH), an enzyme that is released when cells are damaged. Studies have shown that at lower concentrations of curcumin, there was no significant change in ROS, but an increase in the activity of antioxidant enzymes like superoxide dismutase (SOD) and catalase (CAT) was observed. Conversely, higher concentrations of curcumin resulted in a rise in ROS and a reduction in antioxidant enzyme activity, along with significant upregulation of SIRT1, a protein involved in cellular stress responses. Curcumin also regulated fibrogenesis by lowering the levels of fibrotic indicators in treated cells, suggesting its potential role in modulating fibrosis associated with cancer. By modulating oxidative stress, fibrosis, and apoptosis pathways, this study sheds light on how curcumin could overcome drug resistance in cancer therapy [41]. Researchers synthesized various formulations of nanoceria and dextran-nanoceria with curcumin, all demonstrating anti-cancer properties. The nanoparticle delivery of curcumin could be a promising alternative to traditional small-molecule drug therapies for aggressive cancers like neuroblastoma. This approach is advantageous due to its ability to selectively induce apoptosis in cancer cells while minimizing toxicity to healthy cells. When curcumin is delivered by nanoparticles, it changes the Bcl-2/Bax ratio in human neuroblastoma cells, which leads to cellular hypoxia and ROS-mediated apoptosis [42]. Curcumin stimulates apoptosis, particularly through the caspase-8 pathway, and can induce apoptosis without p53's inhibition, a common inhibition in intrinsic apoptotic mechanisms, especially in colon cancers [43]. Researchers demonstrated that curcumin suppresses the growth of colorectal cancer (CRC) cells by targeting various cellular mechanisms and signaling pathways. For instance, it suppresses the activity of cyclin-dependent kinase 2 (CDK2), which is essential for cell cycle progression [44]. Additionally, researchers tested curcumin's antiproliferative effects on HCT-116 and LoVo

colorectal cell lines, which led to a significant reduction in cell viability [45]. Curcumin has been shown to modulate the expression and activity of oncogenic and tumor-suppressive miRNAs, small ncRNAs that regulate gene expression post-transcriptionally and play a role in tumor formation and progression. For example, curcumin can downregulate oncogenic miRNAs and upregulate tumor-suppressive miRNAs, contributing to its potential anti-cancer effects [46] (Fig. 1).

LncRNAs biogenesis, characterization, and function

LncRNAs have a complex and varied structure, often undergoing splicing, polyadenylation, and capping like mRNAs. Their structure can include multiple exons and introns, and they can form secondary structures such as loops and stems that are crucial for their function. Some lncRNAs have a modular structure and contain repeats, which can be relevant to their function [47]. RNA polymerase II transcribes linear lncRNAs, a process similar to that of protein-coding genes. RNA polymerase II typically caps them with 7-methylguanosine at their 5' end, splices them, and polyadenylates them at their 3' end. The splicing of lncRNAs can occur co-transcriptionally, meaning it happens simultaneously with transcription. However, lncRNAs typically splice less efficiently than mRNAs, and polyadenylation signals often do not influence their transcription termination [48, 49].

The different end structures of linear lncRNAs appear to preserve their core sequences while delivering localized signals [50]. LncRNAs are characterized by their chromosomal positions relative to nearby protein-coding genes into sense, antisense, bidirectional, intronic, and intergenic lncRNAs [51]. Linear lncRNAs have diverse functions, including maintaining nuclear structure integrity, regulating genes positively or negatively in cis or trans by recruiting transcription factors or chromatin-modifying complexes to DNA targets in the nucleus, acting as decoys to sequester RNA-binding proteins (RBPs), or directly binding to DNA [52, 53]. In the cytoplasm, linear lncRNAs function as competing endogenous RNAs (ceRNAs) for miRNAs. Additionally, linear lncRNAs can influence signaling pathways by interacting with RBPs [54]. Although linear lncRNAs do not conduct sequence-dependent activities, their loci are frequently used to generate regulators such as enhancers and promoters. The transcription of linear lncRNAs can affect the expression of neighboring genes by attracting particular protein factors [55]. The expression of lncRNAs is tissue-specific and is especially abundant in the brain [53]. The functions of lncRNAs are as diverse as their structures and regulate the expression of target gene at transcriptional and post-transcriptional levels. LncRNAs may function as scaffolds for protein complexes, influence chromatin remodeling, and affect mRNA stability and translation. They play roles in various biological processes such as development,

differentiation, and disease pathogenesis, and can interact with DNA, RNA, and proteins to carry out their functions [56]. LncRNAs, especially Xist (X inactive-specific transcript), play a role in gene silencing on the X-chromosome, especially during X-chromosome inactivation (XCI). Other lncRNAs, such as Jpx and Tsix, also play significant roles in regulating XCI. Jpx activates Xist, while Tsix acts as an antisense RNA that represses Xist on the active X chromosome. This interaction adds a layer of regulation to the XCI process [57, 58].

LncRNAs can affect intracellular signaling pathways, which are essential for cell communication and function. They modulate these pathways by interacting with various components, such as receptors, enzymes, and other signaling molecules [59]. LncRNAs can function as decoys by binding to and sequestering proteins away from their targets, thereby affecting the signaling cascade. They can also serve as scaffolds, assembling multiple proteins into a complex to either facilitate or inhibit signal transduction [60]. In colon cancer, overexpression of the lncRNA DMTF1v4 enhances cell

migration via the ERK/MAPK signaling pathway. Similarly, lncRNA PVT1 promotes cell migration and invasion by modulating miR-26b and miR-30d-5p, which in turn affects downstream signaling in colon cancer cells [61]. By regulating protein-coding genes, lncRNAs can influence signaling pathways crucial for embryogenesis, organ differentiation, and disease progression, including cancer. LncRNAs have been found to regulate every step of gene expression, and their abnormal expression is associated with various cancer types, contributing to both their development and metastatic spread [59, 62]. Certain lncRNAs act as promoters or inhibitors of metastasis, regulating the process at various levels, including transcription, translation, epigenetic modifications, and signaling pathways [63] (Fig. 2).

CircRNA biogenesis, characterization, and function

CircRNAs are small peptides produced by specific functional linear lncRNAs [64]. CircRNAs are more resistant to most RNA degradation processes than linear ncRNAs due to

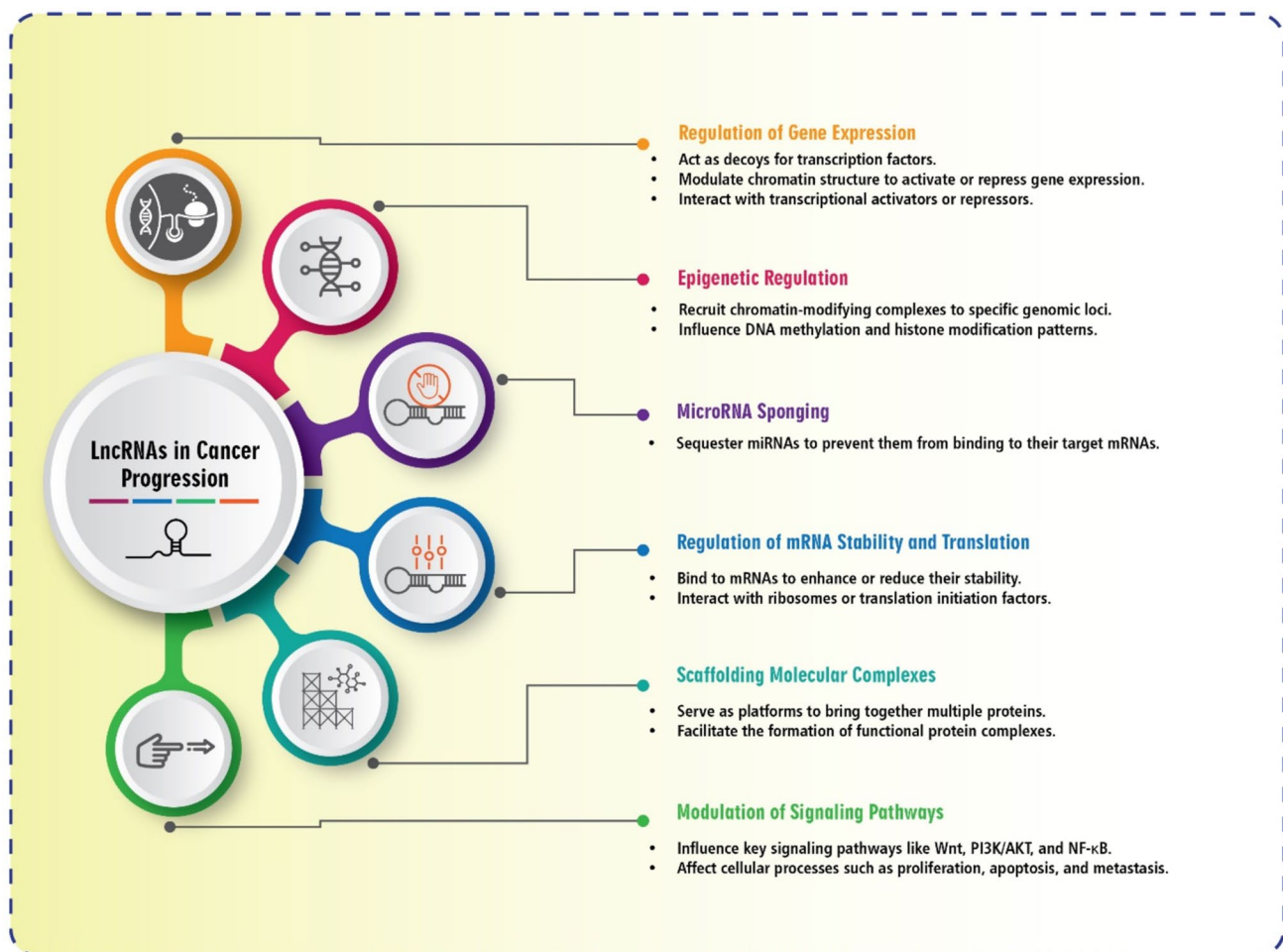


Fig. 2 The biological functions of lncRNAs in cancer regulation

their stability [65]. CircRNAs form a covalently closed continuous loop, unlike the linear structure of most RNAs. Their biogenesis involves the transcription of a gene to produce a linear precursor mRNA (pre-mRNA). During splicing, specific RBPs and the spliceosome machinery facilitate the back-splicing event that joins the 3' and 5' ends of the pre-mRNA to form a circular structure. The resulting circRNA undergoes further processing, such as capping and polyadenylation, to become a mature, stable circular molecule [66, 67]. CircRNAs perform various biological functions, including miRNA sponging, transcriptional regulation, translation, and cellular signaling. CircRNAs can act as “sponges” by binding to and sequestering microRNAs (miRNAs), making them less available to target mRNAs and changing gene expression. Some circRNAs can interact with and sequester specific proteins, affecting their cellular localization or activity. Additionally, certain circRNAs bind to transcription factors or chromatin-modifying compounds to control the expression of their precursor genes. Despite the non-coding nature of most circRNAs, some can translate into functional proteins, contributing to the cellular proteome. CircRNAs can also act as signaling molecules, transmitting information between cells or within the same cell [68]. Researchers created a circRNA modular cloning system that allows for the evaluation of many possible sequence variants and the independent adjustment of several variables. The study systematically dissected five factors that control circRNA translation: vector topology, 5' and 3' untranslated regions, the internal ribosome entry sites, and synthetic aptamers. Optimizing these elements led to a several 100-fold improvements in circRNA protein yields. Improved circRNAs outperformed mRNAs in terms of protein levels in vitro and demonstrated longer-lasting translation and functionality in mice. This research marks a significant advancement in the therapeutic use of circRNAs, providing insights into maximizing circRNA translation for better protein yields, which is crucial for their application in medicine [69].

CircRNAs have the ability to regulate angiogenesis through their function as miRNA sponges. By sequestering miRNAs, they prevent these small regulatory RNAs from inhibiting their target mRNAs involved in angiogenesis. Researchers have reported that abnormal circRNA expression impacts angiogenesis, thereby influencing tumor development. These effects align with various tumorigenic features, known as the “hallmarks of cancer,” including sustained angiogenesis [70]. Some circRNAs have been found to regulate genes directly involved in the cell cycle, apoptosis, and differentiation, processes often dysregulated in leukemia. For instance, a novel circRNA, hsa_circ_0000370, was shown to increase cell survival and inhibit apoptosis in FLT3-ITD-positive acute myeloid leukemia cells by regulating miR-1299 and S100A7A [71]. CircRNA vaccines are based on circRNAs, naturally occurring molecules in

cells with a unique circular structure that provides greater stability compared to linear RNAs. This makes them an attractive platform for vaccine development. One of the key advantages of circRNA vaccines is their enhanced stability. Because they are circular and lack free ends, they are less susceptible to degradation by cellular enzymes, allowing for potentially longer-lasting presence in the body and a more sustained immune response.

CircRNA vaccines work by encoding antigens that, once expressed in the body, stimulate an immune response. This response can be tailored to target specific pathogens or disease-causing agents. CircRNA molecules can be designed to incorporate internal ribosome entry sites (IRESs) and open reading frames (ORFs), essential for the translation of the encoded antigen. The use of circRNA vaccines extends beyond infectious diseases and holds promise for treating non-infectious diseases, such as cancer. By encoding tumor-specific antigens, circRNA vaccines could help the immune system recognize and attack cancer cells [72] (Fig. 3).

Effects of curcumin on circRNA and lncRNA expression in Cancer

Researchers have conducted several investigations to explore the relationship between curcumin's effects and the alteration of circRNA and lncRNA levels. The next sections provide examples of curcumin treatment's ability to change lncRNA (Fig. 4 and Table 1) and circRNA (Fig. 5 and Table 2) expression in different malignancies.

Lung cancers

lncRNAs and circRNAs have been shown to regulate cancer cell growth, migration, autophagy, apoptosis, stemness, tumor microenvironment, and drug resistance, thereby contributing to lung cancer development. CircRUNX1 is a newly discovered functional circRNA that is overexpressed in multiple cancers, including CRC [73], papillary thyroid cancer [74] and lung cancer [75]. It has been observed that curcumin significantly increased the expression of miR-760, acting as a sponge for circRUNX1 in lung cancer cells. This action inhibited circRUNX1 function and subsequently decreased the expression of RAB3D. Upregulation of RAB3D has been implicated in several cancers, including CRC, esophageal squamous cell carcinoma, and lung cancer [76–78]. As a result, curcumin demonstrates an anticancer role in lung cancer cells by influencing the circRUNX1/miR-760/RAB3D axis. This contributes to decreasing Bcl-2, an anti-apoptotic marker, and increasing Bax, an apoptosis marker, thereby enhancing the rate of apoptosis among tumor cells. Moreover, curcumin inhibits the migration and invasion of cancer cells [79].



Fig. 3 The biological functions of circRNAs in cancer regulation

Recent research has demonstrated that the overexpression of circ-PRKCA has a tumorigenic effect in non-small cell lung cancer (NSCLC) [80]. MiR-384 can act as a sponge for circ-PRKCA, and several studies have indicated that it plays a repressive role in various tumors, including osteosarcoma, breast cancer, CRC, and laryngeal cancer [81–85]. Researchers have reported that curcumin treatment in NSCLC enhances the expression of miR-384 while decreasing the expression of circ-PRKCA. These changes facilitate lower the expression of ITGB1 (CD29) expression, increase apoptosis, and reduce invasion and metastasis of cancer cells [81]. ITGB1 is an essential factor for vasculogenesis in cancer development, including prostate cancer and CRC [86–88]. Therefore, curcumin may inhibit NSCLC by regulating the circ-PRKCA/miR-384/ITGB1 axis.

In addition to circRNAs, overexpression of several lncRNAs such as lncRNA-UCA1 has been observed in multiple cancers, including bladder cancer, gastric cancer, CRC, and lung cancer [89–92]. Previous studies have indicated that lncRNA-UCA1 can reduce apoptosis by decreasing the

levels of the proapoptotic agents caspase-3 and caspase-9 [93, 94]. Wang et al. found that curcumin may lower the expression of lncRNA-UCA1 in lung cancer, subsequently inhibiting cancer colony formation and promoting cancer cell apoptosis by enhancing apoptosis-associated factors (caspase-3, -9). Furthermore, curcumin could suppress the Wnt and mTOR pathways—two crucial signaling pathways in human cancer progression—by decreasing the expression of Wnt3a and β -catenin and inhibiting the phosphorylation of mTOR [95]. Furthermore, it has been revealed that lncRNA-MEG3 is expressed in normal tissue and exerts its anti-tumor function by inhibiting proliferation and inducing apoptosis [96–99]. Furthermore, reports suggest that the downregulation of lncRNA-MEG3 can affect the process of chemotherapeutic resistance in lung cancer through various signaling pathways [100, 101]. Clinical studies have further demonstrated the downregulation of phosphatase and tensin homolog (PTEN) in lung cancer. PTEN, a crucial member of the PTP enzyme family, plays a crucial role in cancer suppression by controlling several functional activities in cancer

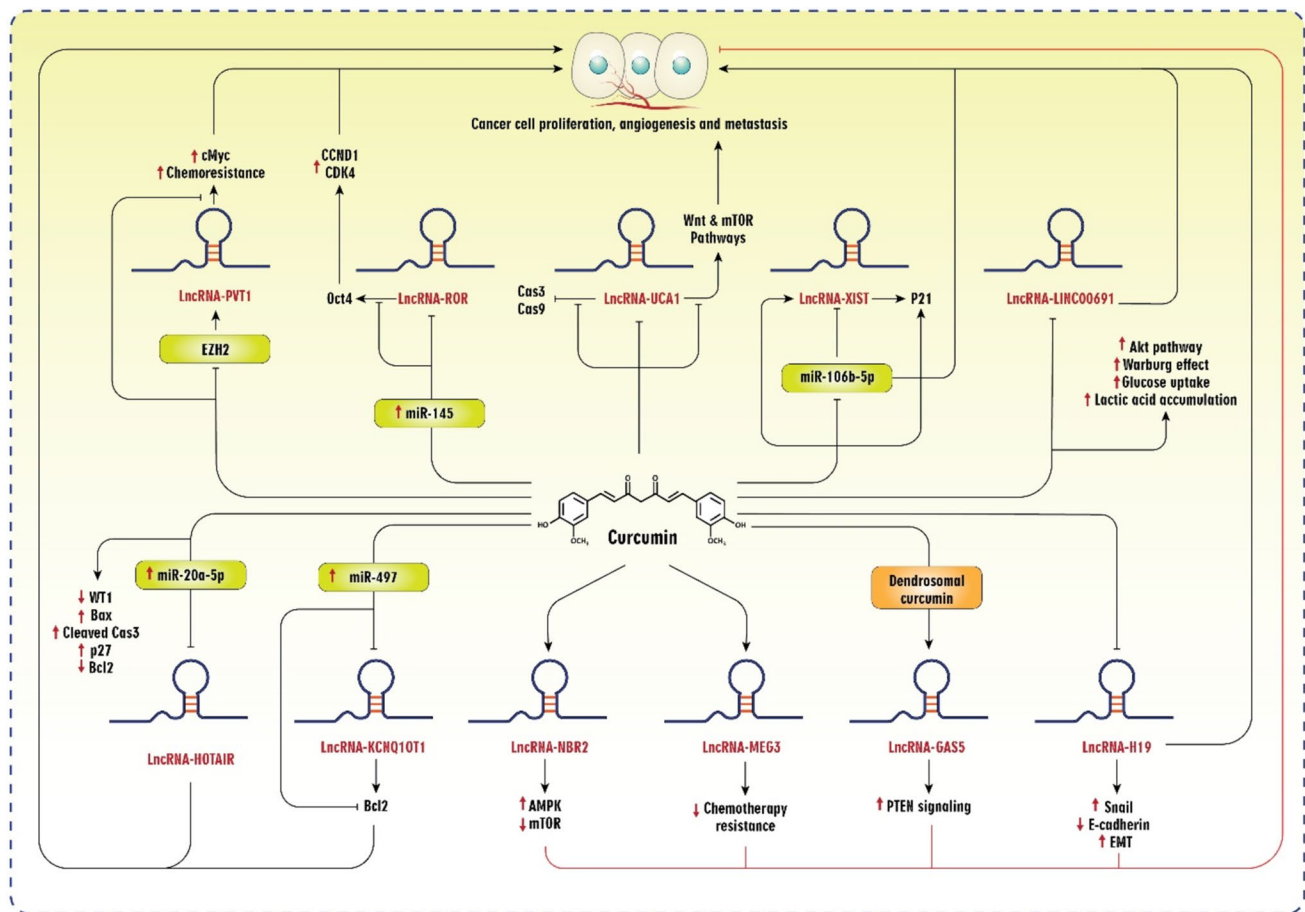


Fig. 4 The role of curcumin on the lncRNA alterations in various cancers

cells, including proliferation, apoptosis, angiogenesis, and metastasis [102]. Curcumin can upregulate the expression of MEG3 and enhance PTEN signaling. As a result, curcumin may suppress lung cancer through the regulation of lncRNA-MEG3 and PTEN [103]. In conclusion, curcumin controls lung cancer by enhancing the cancer cells apoptosis ratio, hindering angiogenesis and vascularization in tumor tissue, and suppressing the proliferation, invasion and migration of cancer cells.

Genitourinary tumors

Prostate cancer

ncRNAs, including lncRNAs, have been identified as important regulators of gene transcription and signal transmission in the pathogenesis of prostate cancer. Some studies have shown that lncRNA-ROR plays a significant role in protecting cancer stem cells (CSCs) and facilitating tumor development in several malignancies such as breast, osteosarcoma, ovarian, gastric, and prostate cancers [104–107].

lncRNA-ROR enhances the expression of Octamer-binding transcription factor (Oct4). Oct4 plays a crucial role in regulating CSCs and is important in tumorigenesis and therapy resistance [108, 109]. Oct4 can promote cancer cell proliferation and tumor progression by increasing the levels of cell cycle proteins such as CCND1 and CDK4 [110, 111]. Liu et al. demonstrated that curcumin treatment in prostate cancer cells increased miR-145 expression. Because miR-145 binds to lncRNA-ROR, increasing miR-145 caused lncRNA-ROR to be silenced, which in turn caused Oct4 to be downregulated. Collectively, curcumin-induced cell cycle arrest and inhibited cancer cell progression and migration via the lncRNA-ROR/miR-145/Oct4 axis [112].

Renal cell carcinoma

Several studies have demonstrated the functional role of lncRNA-XIST in various human cancers, including pancreatic cancer, lung cancer, breast cancer, gastric cancer, cervical cancer, and renal cell carcinoma (RCC) [113–118]. Low lncRNA-XIST expression has been observed in RCC, attributed to the sponging of two miRNAs, miR-106b-5p and

Table 1 Regulation of lncRNAs expression by curcumin in different types of cancer

Cancer type	lncRNA name	Curcumin effect on lncRNA	Corresponding signaling/protein	Outcome	References
Pancreatic ductal adenocarcinoma (PDAC)	LncRNA-PVT1	Downregulation	EZH2-PVT1-c-Myc axis	Enhancing chemotherapy sensitivity, suppressing cancer cell proliferation, and inducing apoptosis	[163]
Prostate cancer	LncRNA-ROR	Downregulation	LncRNA-ROR/miR-145/Oot4 axis	Suppressing cancer cell proliferation and invasion	[112]
Lung cancer	LncRNA-UCA1	Downregulation	LncRNA-UCA1, Wnt pathway and mTOR pathway	Apoptosis induction via enhancing caspase-3 and caspase-9	[95]
Renal cell carcinoma (RCC)	LncRNA-XIST	Upregulation	LncRNA-XIST/miR-106b-5p/P21 axis	Provoking cell cycle arrest in G0/G1 and inhibiting proliferation	[120]
Papillary thyroid carcinoma (PTC)	LncRNA-LINC00691	Downregulation	LncRNA-LINC00691/p-Akt signaling/Warburg effect axis	Apoptosis induction via Warburg effect inhibition and cancer cell proliferation suppression	[178]
Acute myeloid leukemia (AML)	LncRNA-HOTAIR	Downregulation	LncRNA-HOTAIR/miR-20a-5p/WT1 axis	Suppressing proliferation, promoting apoptosis and increasing chemotherapy sensitivity	[185]
Colorectal cancer	LncRNA-KCNQ1OT1	Downregulation	LncRNA-KCNQ1OT1/miR-497/Bcl-2 axis	Suppressing proliferation, promoting apoptosis and increasing chemotherapy sensitivity	[140]
Colorectal cancer	LncRNA-NBR2	Upregulation	LncRNA-NBR2/AMPK/mTOR axis	Suppressing proliferation and cell cycle	[146]
Lung cancer	LncRNA-MEG3	Upregulation	LncRNA-MEG3 and PTEN signaling	Suppressing proliferation, angiogenesis, invasion and metastasis Induction of apoptosis	[103]
Breast cancer	LncRNA-GAS5	Upregulation	–	Suppressing proliferation, invasion and metastasis Inducing apoptosis and chemotherapy sensitization	[190]
Breast cancer	LncRNA-H19	Downregulation	LncRNA-H19/Snail/E-cadherin axis	Suppressing migration and invasion and EMT of cancer cells	[196]

miR-302c, which regulate tumor development [119]. Sun et al. found that overexpression of lncRNA-XIST reduced cell survival and arrested the cell cycle in the G0/G1 phase by enhancing P21, a cell cycle regulator, through the sponging of miR-106b-5p. Additionally, they discovered that curcumin treatment could increase P21 expression by inhibiting miR-106b-5p function and upregulating lncRNA-XIST. As a result, curcumin may suggest a novel therapeutic strategy for RCC by causing cell cycle arrest through the lncRNA-XIST/miR-106b-5p/P21 axis [120].

Gastrointestinal tumors

Numerous circRNAs influence tumor development, expansion, invasion, and drug resistance in gastrointestinal malignancies such as colorectal cancer (CRC), gastric cancer (GC), liver cancer, HCC, and pancreatic ductal adenocarcinoma.

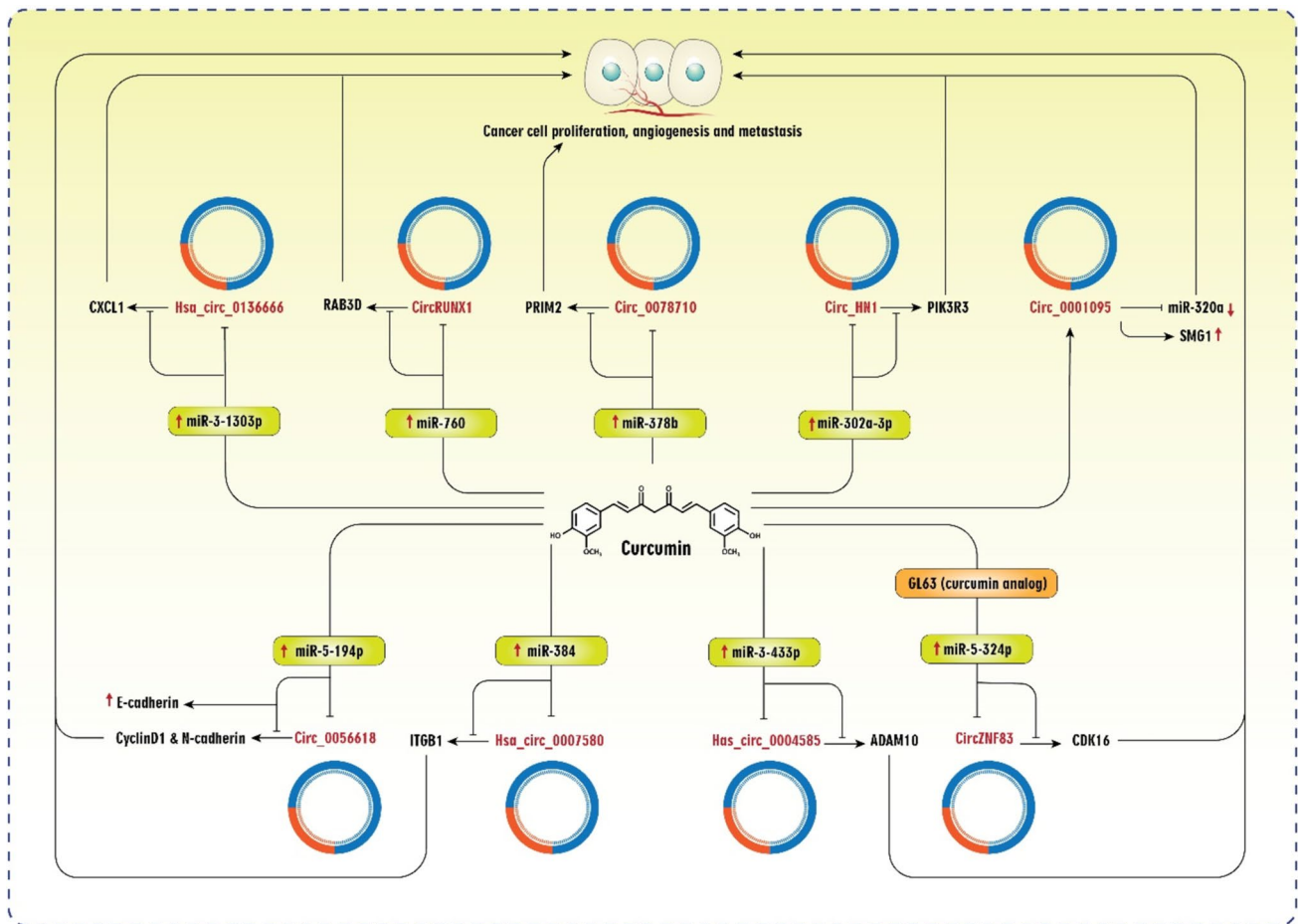


Fig. 5 The role of curcumin in circRNA alterations across various cancers

Colorectal cancer

Hsa_circ_0136666 is recognized as a carcinogenic factor in various tumors, including breast cancer, osteosarcoma, and CRC [46, 121, 122]. Hsa_circ_0136666 accelerates cancer cell proliferation and metastasis in CRC by inducing the expression of CXCL1 [123]. CXCL1 is a chemokine that promotes cancer progression and invasion in multiple cancers, including CRC [124]. MiR-1303-3p can reduce the expression of CXCL1 by interacting with hsa_circ_0136666. It has been shown that curcumin enhances the level of miR-1303-3p, which subsequently decreases CXCL1 expression by repressing hsa_circ_0136666. Furthermore, curcumin treatment significantly reduces Bcl-2, while substantially increasing Bax [125]. Some studies have indicated that circ_HN1 is elevated in various tumors, including gastric cancer, NSCLC, and CRC and this elevation has been linked to TNM staging, tumor size, and lymph node metastasis [21, 61, 126]. Curcumin reduced the oncogenic effects of circ_HN1 overexpression in CRC by increasing the expression of miR-302a-3p, thereby inhibiting phosphoinositide-3-kinase

regulatory subunit 3 (PIK3R3) and mitigating malignant behaviors in CRC cells [127]. MiR-302a-3p can act as a sponge for circ_HN1 and has been observed to be decreased in lung cancer [128]. Moreover, curcumin attenuated the expression of PIK3R3. In summary, curcumin could suppress CRC development, metastasis, and invasion by regulating the circ_HN1/miR-302a-3p/PIK3R3 axis [127]. In addition, circ-KIAA1199 is upregulated in CRC tissue and is considered a diagnostic and prognostic marker for CRC [129]. Circ-KIAA1199 exerts its function by sponging miR-433-3p in CRC, suppressing its expression. A disintegrin and metalloproteinase 10 (ADAM10) is miR-433-3p's target protein. The overexpression of circ-KIAA1199 leads to the downregulation of miR-433-3p, thereby enhancing the expression of ADAM10 in CRC [130]. ADAM10 is a member of the ADAM family with proteolytic activity, contributing to various cellular processes such as cell signal transduction and extracellular matrix (ECM) degradation. Studies have confirmed significant activities of ADAM10 in the growth, apoptosis, invasion, and metastasis of cancers, including CRC [131–133]. Researchers have observed that

Table 2 Regulation of circRNAs expression by curcumin in different types of cancer

Cancer type	circRNA name	Curcumin Effect on circRNA	Corresponding signaling/protein	Outcome	References
Colorectal cancer	Hsa_circ_0136666	Downregulation	Hsa_circ_0136666/miR-1301-3p/CXCL1	Attenuated cancer cell growth and increased apoptosis in tumor cells	[125]
Lung cancer	CircRUNX1 (Circ_0002360)	Downregulation	CircRUNX1/miR-760/RAB3D	Enhanced apoptosis rate in cancer cells and inhibited migration and invasion	[79]
Liver cancer	Circ_0078710	Downregulation	Circ_0078710/miR-378b/PRIM2	Restrained HCC proliferation, metastasis and invasion	[149]
Colorectal cancer	Circ_HN1	Downregulation	Circ-HN1/miR-302A-3P/PIK3R3	Restrained cancer cell proliferation, metastasis and invasion	[127]
Ovarian cancer	Circ_0001095 (Circ-PLEKHM3)	Upregulation	Circ-PLEKHM3/miR-320a/SMG1 axis	Suppressed ovarian cancer cell proliferation and induced apoptosis	[164]
Gastric cancer	Circ_0056618	Downregulation	Circ_0056618/miR-194-5p axis	Inhibiting cancer cells proliferation, migration, invasion and inducing apoptosis	[156]
Non-small cell lung cancer	Hsa_circ_0007580 (circ-PRKCA)	Downregulation	Circ-PRKCA/miR-384/ITGB1 axis	Elevating apoptosis ratio, restraining invasion and metastasis, impeding vasculogenesis	[81]
Colorectal cancer	Hsa_circ_0004585 (circ-KIAA1199)	Downregulation	Circ-KIAA1199/miR-433-3p/ADAM10 axis	Inhibiting cancer cells proliferation, migration, invasion and inducing apoptosis	[130]
Hepatocellular cancer	CircZNF83 (hsa_circ_0052112)	Downregulation	CircZNF83/miR-324-5p/CDK16 axis	Suppressing proliferation promoting apoptosis	[153]

curcumin downregulates the expression of circ-KIAA1199 and ADAM10, while upregulating the expression of miR-433-3p. Therefore, curcumin can modulate CRC cell proliferation, apoptosis, invasion, and metastasis through the circ-KIAA1199/miR-433-3p/ADAM10 axis [130].

Furthermore, several lncRNAs, such as lncRNA-KCNQ1OT1, are associated with several malignancies, including CRC, bladder, lung, and prostate [134–137]. Additionally, lncRNA-KCNQ1OT1 can contribute to the development of cisplatin resistance in tumor cells [138, 139]. It has been shown that lncRNA-KCNQ1OT1 promotes proliferation, inhibits apoptosis, and enhances cisplatin resistance in cancer cells by sponging miR-497, thereby reducing its suppressive effect on Bcl-2 expression [140]. Researchers have reported that Bcl-2 contributes to increased cisplatin resistance in various cancers [141]. Curcumin treatment in CRC can downregulate the expression of lncRNA-KCNQ1OT1 and reduce the expression of Bcl-2 by enhancing the function of miR-497. Therefore, curcumin might be considered a novel therapeutic approach for CRC by regulating the lncRNA-KCNQ1OT1/miR-497/Bcl-2 axis [140]. In

addition, lncRNA-NBR2 is reported as a tumor suppressor in various cancers, including CRC, thyroid cancer, HCC, and breast cancer [142–145]. Glucose deficiency promotes the expression of lncRNA-NBR2 in CRC via adenosine monophosphate-activated protein kinase (AMPK). Upregulation of lncRNA-NBR2 inhibits the activation of the mTOR signaling pathway while enhancing AMPK activation through a positive feedback loop. Altogether, these biological processes lead to the inhibition of cell proliferation [146]. AMPK is considered a tumor suppressor due to its ability to inhibit cell division through metabolic reprogramming and by suppressing the cell cycle regulator, mTOR signaling [147]. Curcumin significantly increased lncRNA-NBR2 expression in CRC cells, leading to AMPK signaling activation and mTOR inhibition. As a result, curcumin could suppress CRC cell proliferation and disrupt the cell cycle through the lncRNA-NBR2/AMPK/mTOR axis [146]. Consequently, Curcumin could modify CRC by promoting apoptosis, suppressing cell proliferation and increasing chemotherapy sensitivity among the cancer cells.

Liver cancer

Studies have reported a significant association between overexpression of circ-0078710 and liver cancer cell proliferation, migration, and invasion [148]. Curcumin decreased the expression of circ-0078710 and concurrently increased the expression of miR-378b. This caused curcumin to lower the level of PRIM2 in hepatocellular cells by disrupting the function of circ_0078710, which is a sponging miR-378b [149]. MiR-378b has been observed to inhibit circ_0078710, which in turn reduces the activation of hepatic stellate cells and mitigating liver fibrosis [150]. DNA primase polypeptide 2 (PRIM2) is a type of DNA primer enzyme that contributes to initiating DNA replication, cell cycle progression, and cell proliferation [151]. In summary, curcumin regulates the proliferation of liver cells and inhibits cell cycle progression in hepatocellular cells. Furthermore, curcumin upregulates the pro-apoptotic protein Bax and downregulates motility-associated proteins such as vimentin and MMP9, restricting the proliferation, metastasis and invasion of liver cancer cells [149].

Hepatocellular carcinoma (HCC)

CircZNF83 upregulation has been observed to increase the proliferation, migration, and invasion of tumor cells while decreasing apoptosis in breast cancer and HCC [21, 152]. CircZNF83 could sponge miR-324-5p and upregulate cyclin-dependent kinase 16 (CDK16). The circZNF83/miR-324-5p/CDK16 axis can regulate the JAK2/STAT3 signaling pathway, which is a key pathway in HCC progression [153]. The JAK2/STAT3 signaling pathway is involved in regulating HCC cell proliferation and apoptosis [154, 155]. GL63, a curcumin analog, can downregulate circZNF83 while upregulating miR-324 in GL63-treated HCC. GL63 blocks the JAK2/STAT3 signaling pathway through the circ-ZNF83/miR-324-5p/CDK16 axis, thereby inhibiting HCC cell growth and induce apoptosis [153].

Gastric cancer

Studies found that circ_0056618 expression level is increased, while miR-194-5p expression is decreased in gastric cancer. Curcumin treatment can elevate the level of miR-194-5p and suppress the expression of circ_0056618 in gastric cancer tissue. Additionally, miR-194-5p acts as a sponge for circ_0056618, inhibiting its function. Curcumin treatment can suppress gastric cancer cell proliferation, progression, and invasion, as well as induce apoptosis, by reducing the protein levels of cyclinD1 and N-cadherin and increasing the protein level of E-cadherin through the regulation of the circ_0056618/miR-194-5p axis [156].

Pancreatic ductal adenocarcinoma

Pancreatic ductal adenocarcinoma (PDAC) cells overexpress Enhancer of Zeste Homolog-2 (EZH2), a catalytic subunit of polycomb repressive complex 2 (PRC2), a histone methyltransferase that plays an oncogenic role. Overexpression of EZH2 increases resistance to gemcitabine during PDAC chemotherapy [157–159]. EZH2 acts as an oncogene by increasing the expression of lncRNA PVT1, which induces gemcitabine resistance in PDAC [160]. PVT1 induces chemoresistance by increasing and stabilizing the c-Myc oncogene. Overexpression of c-Myc occurs in various types of cancer [161, 162]. Curcumin treatment has been shown to enhance the cellular toxicity of gemcitabine during chemotherapy and suppress CSCs by downregulating the EZH2-PVT1-c-Myc axis. Curcumin inhibited the colonization of cancer cells and induced apoptosis by suppressing proliferating cell nuclear antigen (PCNA) while upregulating the cell cycle-regulating tumor suppressor p21 [163].

Other tumors

Ovarian cancer

The expression of various circRNAs has been demonstrated to alter ovarian cancer cell proliferation, invasion, and dissemination. It has been determined that circ-PLEKHM3 expression is reduced in ovarian cancer, contributing to increased cancer cell proliferation and metastasis [21]. Induction of circ-PLEKHM3 expression in cancer tissue significantly increased the levels of c-caspase-3 and Bax, while decreasing the level of PCNA [164]. Additionally, circ-PLEKHM3 acts as a sponge for miR-320a, thereby inhibiting ovarian cancer cell progression [165]. Several studies have revealed that miR-320a functions as an oncogenic miRNA in various tumors, including retinoblastoma, prostate cancer, and ovarian cancer [166–168]. MiR-320a negatively regulates the expression of Suppressor of Morphogenesis in Genitalia 1 (SMG1) [164]. SMG1, a member of the phosphoinositide 3-kinase-related kinase family, exhibits tumor suppressive properties by inhibition of cancer cell growth, invasion, and metastasis in various cancers, including ovarian cancer, acute myeloid leukemia, gastric cancer, hepatocellular carcinoma, and nasopharyngeal carcinoma [169–173]. Sun et al. found that curcumin increases circ-PLEKHM3 expression, which subsequently suppresses miR-320a function and enhances SMG1 levels. Collectively, curcumin can suppress cancer cell proliferation and induce cancer cell apoptosis by regulating the circ-PLEKHM3/miR-320a/SMG1 axis [164]. Furthermore, curcumin can inhibit ovarian cancer cell progression and promote apoptosis by

regulating the AKT/mTOR/p70S6 pathway and blocking the Wnt/ β -catenin pathway [174, 175].

Papillary thyroid carcinoma (PTC)

The research of lncRNAs in PTC is quickly expanding, and new studies have provided insight into their role in the disease's formation and progression. Overexpression of lncRNA-LINC00691 has been observed in several cancers, including gastric cancer and lung cancer [176, 177]. Li et al. indicated that overexpression of lncRNA-LINC00691 is characteristic of PTC. Inhibition of LINC00691 can suppress proliferation and induce apoptosis by enhancing the expression of pro-apoptotic proteins Bad and cleaved caspase-3 while reducing the expression of Bcl-2 in tumor cells. Curcumin treatment can lower the level of LINC00691, leading to suppression of the p-Akt signaling pathway and inhibition of the Warburg effect. This inhibition reduces glucose uptake and lactic acid accumulation in cancer cells [178]. The Warburg effect is a significant metabolic process associated with the occurrence and development of cancer. It inhibits apoptosis in cancer cells by preventing mitochondrial ATP production [179, 180].

Acute myeloid leukemia (AML)

Recent research suggests that lncRNAs such as Lnc-HOTAIR are overexpressed in multiple cancers, including liver cancer, breast cancer, cervical cancer, and AML [181–184]. Overexpression of HOTAIR promotes AML cell proliferation and inhibits apoptosis. Additionally, HOTAIR expression can increase AML cell resistance to Adriamycin. Curcumin can disrupt the expression and function of HOTAIR by promoting miR-20a-5p expression, which acts as a sponge for HOTAIR. Consequently, curcumin reduces the expression of Wilms' tumor 1 (WT1), leading to increased levels of Bax, cleaved caspase-3, and p27 while decreasing the expression of Bcl-2. This enhances the sensitivity of AML cells to Adriamycin [185]. The WT1 gene is a crucial factor contributing to the pathogenesis of several cancers, including AML. It has been observed that silencing WT1 significantly decreases the proliferation of AML cells [186, 187]. Collectively, curcumin treatment in AML may have antimalignant potential by suppressing leukemia cell proliferation and promoting apoptosis through the lncRNA-HOTAIR/miR-20a-5p/WT1 axis [185].

Breast cancer (BC)

Several studies suggest that lncRNAs play an important role in breast cancer carcinogenesis by modulating the expression of tumor-related genes. Dysregulation of the lncRNA growth arrest-specific transcript 5 (GAS5) is associated with various

types of cancer, including breast cancer [188]. Dendrosomal curcumin (DNC) is a nanoparticle formulation of curcumin that is safe and non-toxic to normal cells. It shows promise in cancer treatment due to its increased bioavailability and targeted delivery [189]. DNC increased the expression of lncRNA-GAS5, known as a tumor suppressor gene. Overexpression of GAS5 can reduce chemotherapy resistance and enhance the antitumor effects of curcumin, including anti-proliferative, anti-metastatic, cell cycle arrest, and activation of apoptosis in breast cancer cells. Knocking down GAS5 reduced the apoptotic, antiproliferative, and antimetastatic effects of DNC. Downregulation of GAS5 is a key factor in resistance to DNC treatment in breast cancer cells. By increasing GAS5 expression in conjunction with DNC treatment, it might be possible to improve treatment efficacy and potentially overcome resistance mechanisms that hinder the anticancer effects of DNC in breast cancer [190]. Tamoxifen-resistant (TAMR) breast cancer has a higher probability of invasion and metastasis. lncRNA-H19 plays a significant role in the proliferation, epithelial-mesenchymal transition (EMT), metastasis, chemoresistance, and endocrine resistance of TAMR breast cancer [191, 192]. EMT is crucial for the migration and invasion of tumor cells [193]. Key markers of EMT include E-cadherin and vimentin, with E-cadherin expression decreasing and vimentin expression increasing during the process [194]. Overexpression of H19 leads to an increase in vimentin and a decrease in E-cadherin, thereby promoting migration and invasion. Conversely, downregulation of H19 increases E-cadherin and decreases vimentin, inhibiting migration and invasion. The zinc finger transcriptional repressor snail is a critical regulator of EMT, as it downregulates epithelial genes and upregulates mesenchymal genes [195]. lncRNA-H19, as an upstream regulatory factor, upregulates Snail, which decreases E-cadherin expression and induces EMT in TAMR cells. Researchers have found that curcumin inhibits the lncRNA-H19/Snail/E-cadherin axis, thereby reducing H19-induced migration, invasion, and EMT. Therefore, curcumin may be a promising therapeutic option for treating patients with TAMR breast cancer by inducing apoptosis and enhancing the sensitivity of cancer cells to the chemotherapy [196].

Curcumin's therapeutic potential in cancer

Researchers have conducted several human trials to determine the beneficial implications of curcumin on cancer patients. Studies have demonstrated its ability to alleviate symptoms and enhance tumor markers. Curcumin has been used in capsule form or as an ointment to treat various malignancies, including breast [197], prostate [198], CRC [199], pancreatic [200], endometrial [201], and oral cancers [202]. In most trials, turmeric oil or extract offered therapeutic relief and enhanced the state of histological lesions.

Oral curcumin therapy in myeloma patients reduced para-protein and urinary N-telopeptide of type I collagen levels and downregulated the expression of NF- κ B, STAT3, and COX-2 [203]. Furthermore, curcumin treatment was found to be beneficial for patients with orbital pseudotumors, head and neck squamous carcinoma, and breast, lung, and prostate cancers. It resulted in reductions in tumor size and tumor markers, as well as a decrease in secreted mutagens [204–206]. However, while curcumin improved the quality of life, tumor markers, and serum cytokines, it had a minimal effect on pancreatic cancer patients. The advanced stages of the disease [207–210] could account for this limited efficacy [207–210]. Taking curcumin, mostly in the form of the C3 complex, decreased serum TNF- α levels, reduced the number and size of polyps and aberrant crypt foci, and increased tumor apoptotic cells, as well as the expression of p53, Bax, and Bcl-2 in CRC patients [211–213]. A study of 97 prostate cancer patients found that taking 1.44 g of curcumin daily for 6–36 months reduced the level of prostate-specific antigen [214]. However, curcumin had no substantial effect in several conditions. For example, a randomized controlled trial found that administering nanocurcumin (120 mg/day) had no significant impact on radiation-treated prostate cancer patients [214]. Curcumin (6 g/day) did not significantly improve metastatic castration-resistant prostate cancer after 6 weeks of treatment [215]. The conflicting results may be due to the numerous variables involved in clinical studies, necessitating further investigation.

Conclusion

Natural medicine has long been recognized for its potential to inhibit tumor development and serve as supplementary or stand-alone cancer treatments [18, 116, 117]. Curcumin is an herbal medicine that is commonly utilized for the treatment of various cancers. It works by restricting tumor cell proliferation, inhibiting aggression and migration, inducing apoptosis and autophagy, and reducing cancer stemness. Curcumin can be utilized both alone and in combination with other pharmaceuticals, known for their low toxicity and compatibility [118–121]. ncRNAs have been found to have key functions in tumor cell development, making them potential targets for new anti-cancer therapies. Curcumin also influences the expression of lncRNAs and circRNAs, which affects their miRNA sponging function. Regulating tumor suppressor or oncogene lncRNAs/circRNAs axis could be a promising approach for curcumin's clinical effects.

The clinical application of ncRNA therapeutics presents several challenges during treatment. Firstly, ncRNAs are vulnerable to degradation, necessitating protective modifications to enhance their stability in biological settings.

Besides, the immune system can initiate an inflammatory response upon detecting exogenous RNAs. Moreover, achieving effective delivery of RNA therapeutics to specific target cells is a considerable challenge. It is crucial to reduce undesirable "on-target" effects that can happen when these therapies are absorbed by non-target cells. Additionally, "off-target" interactions may occur due to sequence similarities in RNAs or from using doses that surpass typical endogenous levels. As well, excessive amounts of exogenous RNA can overload the cell's natural RNA processing mechanisms, which may result in reduced therapeutic effectiveness. Finally, the development and production of ncRNA therapeutics can be costly, which affects their accessibility for patients [216–219].

Given the regulatory roles of lncRNAs/circRNAs in cancer formation and progression, as well as their value as cancer-predictive indicators, further research is needed to understand curcumin's impact on the overall expression patterns of lncRNAs/circRNAs in specific tumors. Moreover, most studies on the effects of curcumin on lncRNAs/circRNAs in malignancies have been conducted in vitro, necessitating more research to evaluate the modulatory effects of curcumin and curcumin nanoformulations on lncRNAs/circRNAs expression profiles in preclinical and clinical settings. Addressing several challenges could enhance curcumin's medicinal properties. In clinical practice, curcumin's bioavailability is limited due to fast metabolism, poor absorption and low water solubility. Encapsulation of curcumin in various nanostructures can improve its bioavailability and clinical benefits. Additionally, using hydrophilic nanostructures can overcome curcumin's solubility issues. Encapsulating curcumin in nanoparticles with a strong affinity for biological membranes, such as liposomes, can enhance absorption and uptake. Notably, modifying nanostructures' surfaces for tumor cell targeting can further improve the therapeutic benefits of curcumin. More clinical trials are needed to determine the anticancer effects of curcumin in humans.

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